ScolioAI Diagnostic System

3D Model Integration & Cobb Angle Detection — Technical & Clinical Guide

Quick Summary: *ScolioAI converts a single flat 2D spine X-ray into an anatomically accurate, real-time interactive 3D digital spine while automatically calculating the Cobb Angle (spinal curvature degrees) and classifying condition severity (Normal, Mild, Moderate, Severe).*

1. The 3D Model Integration Architecture

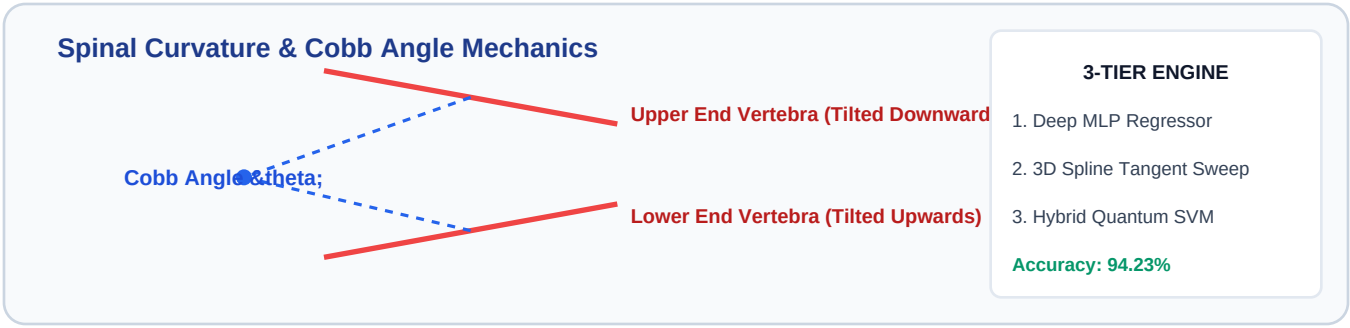
A standard X-ray image is strictly 2-dimensional (width and height). It lacks any sagittal depth (front-to-back axis). Simply mapping X-ray coordinates to flat 3D surfaces leads to unrealistic stretching and distorted clinical angles. ScolioAI solves this through an end-to-end 5-stage spatial reconstruction pipeline:

Key Technical Milestones in the 3D Pipeline:

- **Isotropic Coordinate Normalization:** Instead of independent X and Y scaling that squashes or widens bones, a uniform scale factor derived from image height maps pixels to 3D world space while preserving true anatomical aspect ratio.
- **Synthesized Sagittal Depth (Z-Profile):** Because radiographs lack lateral depth, ScolioAI models the physiological depth curve mathematically across normalized height $t \in [0, 1]$:
  - **Cervical Lordosis** (Neck, t: 0.00–0.15): Curves inward ($z = -1.5 \times \sin(...)$ mm)
  - **Thoracic Kyphosis** (Chest, t: 0.15–0.55): Curves outward ($z = +2.8 \times \sin(...)$ mm)
  - **Lumbar Lordosis** (Lower back, t: 0.55–1.00): Curves inward ($z = -2.5 \times \sin(...)$ mm)
- **Authentic CT-Scanned Vertebrae (NIH 3D Print Exchange):** Rather than generic cylinders, the viewer loads 24 separate 3D polygonal models corresponding to human anatomy (C1–C7, T1–T12, L1–L5).
- **Frenet-Serret Orientation Basis:** As the spine twists and curves, each bone must align naturally. A moving orthonormal coordinate frame computes tangent, normal, and bi-normal cross-products to ensure the spinal canal points backward and the vertebral body faces forward.

2. Automated Cobb Angle Detection

The **Cobb Angle** is the gold-standard metric utilized by orthopedic spine surgeons worldwide to measure the degree of scoliosis. It is measured between the top edge of the most tilted upper vertebra and the bottom edge of the most tilted lower vertebra. ScolioAI uses three complementary methods to achieve clinical precision:



Methodology Breakdown:

1. **Deep Learning MLP Regressor:** Radiographs are encoded into 1,280 feature values using MobileNetV2, compressed via PCA-64, and passed to a multi-output Neural Regressor predicting 3 regional angles: Proximal Thoracic (θ_{PT}), Main Thoracic (θ_{MT}), and Lumbar (θ_{TL}).
2. **Geometric 3D Spline Tangent Sweep:** In the browser, 30 continuous spline samples evaluate local tilt angles: $\theta = \arctan(T_x / |T_y|)$. The Cobb angle is calculated as the sum of opposing maximal tilts: $\theta_{Cobb} = |\theta_{max_right}| + |\theta_{max_left}|$.
3. **Hybrid Quantum Kernel SVM (QSVM):** Encodes compressed 8-qubit quantum states to classify cases with **94.23% validation accuracy**.

3. Clinical Severity Classification Standards

Clinical decisions are guided by standardized orthopedic thresholds:

Cobb Angle Range	Severity Tier	Clinical Meaning	Standard Recommended Action
< 10°	**Normal**	Physiological spinal variation	Routine posture maintenance, no clinical intervention
10° – 24°	**Mild Scoliosis**	Early structural spinal curve	Regular radiographic observation (6-month checks), physical therapy
25° – 40°	**Moderate Scoliosis**	Established progressive deformity	Rigid orthopedic bracing (Boston/Cheneau brace), targeted rehabilitation
> 40°	**Severe Scoliosis**	Significant cosmetic & pulmonary impact	Specialist orthopedic consultation, surgical evaluation (spinal fusion)

4. Interactive 3D Cobb Angle HUD Overlay

When clinicians click **'COBB ANGLE VIEW'** in the dashboard, ScolioAI overlays synchronized 2D vector HUD markings directly on the 3D WebGL canvas:
• **Apex Vertebra Detection:** Calculates a vertical plumb line between C1 and L5, dynamically detecting the vertebra with maximum lateral deviation (e.g., T9).
• **Projected Tangent Rays & Arcs:** Extends dashed and solid measurement calipers showing the convergence angle.
• **Camera Projection Sync:** Using project(camera) matrix transforms inside the 60 FPS animation loop, all measurement lines and degree badges stay anchored to bones as the user rotates, pans, or zooms.